Exact Root Certificates

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From an approximate root to an exact certificate

First edition, 15 October 2026. This is a study and a small exact verifier for a classical sufficient criterion. It makes no claim of mathematical novelty or Lean verification. The executable scope is square systems of polynomials with rational coefficients over the real numbers.

An approximate answer is useful input to a proof, but is not yet a proof. This toolkit turns a rational center, a rational box, and a rational matrix into checkable inequalities which imply that **exactly one real zero lies in that specified closed box**, together with an explicit bound on its distance from the center. A rejected candidate means only that this particular test did not certify it.

1. What the extra information does

Suppose a numerical calculation suggests that $F(c)$ is small. Even an exact small residual cannot on its own establish a nearby root. For example, $f(t) = t^2 + \varepsilon$, with $\varepsilon > 0$, has residual $f(0) = \varepsilon$ and has no real zero. Also $g(t) = \varepsilon(t - 1)$ has residual of magnitude ε at zero, while its root stays distance one away. Multiplying equations by small constants can shrink residuals without improving an approximation.

We need both inverse scale information and control of derivative variation on a whole region. Choose a fixed matrix Y, usually an exact rational approximation to $DF(c)^{-1}$, and consider the **map**

$$T : B \longrightarrow \mathbb{R}^n, \qquad T(x) = x - YF(x).$$

Here T is a function; $T(c) = c - YF(c)$ is one vector. A small value of $YF(c)$ controls a first step, whereas a bound on DT controls the map throughout the box. The matrix Y stays fixed as x varies. This map is generally different from the Newton map $x \mapsto x - DF(x)^{-1}F(x)$.

If Y is invertible, $T(x) = x$ is equivalent to $F(x) = 0$. If it is singular, this implication can fail: $YF(x) = 0$ need not imply $F(x) = 0$. The verifier therefore checks invertibility exactly.

2. The certificate theorem

Let $n \geq 1$, $c \in \mathbb{R}^n$, and $r_i > 0$. Define

$$B = \prod_{i=1}^{n} [c_i - r_i, c_i + r_i], \qquad \|v\|_r = \max_i \frac{|v_i|}{r_i}.$$

Suppose $F : U \to \mathbb{R}^n$ is continuously differentiable on an open set U containing B. Let $Y \in \mathbb{R}^{n \times n}$ be invertible. An interval matrix J is an array of closed real intervals such that

$$\partial_j F_i(x) \in J_{ij} \quad \text{for every } x \in B.$$

Using interval operations, form $A = I - YJ$, and set

$$M_{ij} = \max(|\inf A_{ij}|, |\sup A_{ij}|), \quad d_i = |(YF(c))_i|, \quad s_i = d_i + \sum_j M_{ij} r_j,$$

$$q = \max_i \frac{\sum_j M_{ij} r_j}{r_i}, \qquad \eta = \max_i \frac{d_i}{r_i}.$$

Theorem. If

$$\boxed{s_i \leq r_i \text{ for every } i, \qquad q < 1,} \tag{C}$$

then F has a unique zero ξ in the entire closed box B. Moreover

$$\boxed{\|\xi - c\|_r \leq \min\left(1, \frac{\eta}{1-q}\right), \qquad |\xi_i - c_i| \leq r_i \min\left(1, \frac{\eta}{1-q}\right).} \tag{E}$$

All the quantities can be rational when the inputs and derivative enclosures are rational. There is no appeal to a rounded decimal comparison with one. The componentwise self-map test in (C) should not be replaced by the stronger shortcut $\eta + q \leq 1$: that shortcut is sufficient but may discard valid certificates whose worst residual and derivative bounds occur in different rows.

Proof

First obtain a difference bound valid on the whole box. For any $x, y \in B$, their connecting segment lies in B, since a box is convex. By the fundamental theorem of calculus, component by component,

$$T(x) - T(y) = \int_0^1 (I - YDF(y + t(x - y)))(x - y)\, dt.$$

The interval construction gives $|(I - YDF(z))_{ij}| \le M_{ij}$ for every $z \in B$. Thus

$$|T_i(x) - T_i(y)| \le \sum_j M_{ij}|x_j - y_j|, \qquad \|T(x) - T(y)\|_r \le q\|x - y\|_r. \tag{1}$$

There is generally no single intermediate point for which the familiar scalar mean-value formula becomes a vector identity. The integral avoids that incorrect substitution.

Preserving the domain. Since $T(c) - c = -YF(c)$, for $x \in B$,

$$|T_i(x) - c_i| \le d_i + \sum_j M_{ij}|x_j - c_j| \le s_i \le r_i.$$

Hence $T(B) \subseteq B$. This is the role of the first set of inequalities. It ensures the iteration never leaves the region on which our derivative bounds are valid.

Existence. Put $x_0 = c$ and $x_{k+1} = T(x_k)$. Every iterate lies in B. Equation (1) implies, by induction,

$$\|x_{k+1} - x_k\|_r \le q^k \eta.$$

For $m > k$, the triangle inequality and the geometric series give

$$\|x_m - x_k\|_r \le \eta \sum_{j=k}^{m-1} q^j \le \frac{q^k \eta}{1 - q}.$$

Here $0 \le q < 1$, so the iterates form a Cauchy sequence. The weighted norm is equivalent to the usual norm on $\mathbb{R}^n$; a closed box is complete. They converge to a point $\xi \in B$. Continuity gives $T(\xi) = \lim_k T(x_k) = \lim_k x_{k+1} = \xi$. Invertibility of Y then gives $F(\xi) = 0$.

Uniqueness. If $u, v \in B$ are zeros, both are fixed points. Equation (1) gives $\|u - v\|_r \le q\|u - v\|_r$, which forces $u = v$. This proves uniqueness everywhere in B, including its boundary.

Error. Apply (1) to ξ, c:

$$\|\xi - c\|_r \le \|T(\xi) - T(c)\|_r + \|T(c) - c\|_r \le q\|\xi - c\|_r + \eta.$$

Rearrangement proves the fraction in (E), and $\xi \in B$ supplies the additional bound one. This proves the theorem. The same argument also gives $\|\xi - x_k\|_r \le q^k \eta/(1 - q)$ for its exact iterates. The executable checker certifies the center bound; it does not compute an iteration history.

What each hypothesis contributes

A continuous self-map of a nonempty compact convex box has a fixed point by the Brouwer fixed-point theorem, even without contraction. Thus the self-map inequalities alone, with invertible Y, also suffice for existence via that separate classical theorem. They do not establish uniqueness. The proof above is a direct constructive proof of our combined criterion and does not require Brouwer. The executable reports `certified` only for the combined criterion; it is not a separate existence-only verifier.

Conversely, contraction without a self-map need not give a root in B: $F(t) = t - 2$, $B = [-1, 1]$, $Y = 1$ gives a constant map $T(t) = 2$ with Lipschitz constant zero. Nonstrict self-inclusion is deliberate: for $F(t) = t - 1$ on the same box the root is on the boundary and certification with $q = 0$ is valid. Strict $q < 1$ is also deliberate: $F = 0$ and $Y = 1$ give $q = 1$, a self-map, and infinitely many roots.

The derivative bound also forces $DF(x)$ to be nonsingular at every $x \in B$. If $DF(x)v = 0$ for nonzero v, then $(I - YDF(x))v = v$, contradicting (1)'s matrix bound $\|v\|_r \le q\|v\|_r$. This criterion therefore cannot certify a multiple or singular root, even when that root happens to be locally unique.

3. How intervals make the hypotheses computable

For rational intervals $[a, b]$ and $[u, v]$, use

$$[a, b] + [u, v] = [a + u, b + v], \quad [a, b][u, v] = [\min(au, av, bu, bv), \max(au, av, bu, bv)].$$

Negation reverses and negates the endpoints. A nonnegative integer power is monotone on the real line for odd powers; for positive even powers an interval crossing zero has lower endpoint zero and upper endpoint $\max(a^k, b^k)$. The zeroth power is the constant interval $[1, 1]$, also when the input interval contains zero: this is evaluation of the constant monomial. These rules enclose all possible point values.

A polynomial is supplied as coefficients and exponent vectors. Differentiate each monomial exactly. Substitute the coordinate intervals of B into those derivative polynomials to form J. Induction over additions, multiplications and powers proves the enclosure property required above. Signed multiplication by entries of Y matters when forming A.

The implementation uses Python's arbitrary size integers and `Fraction` for every arithmetic endpoint and comparison. It performs no floating arithmetic in the certificate calculation. Rational endpoints are exact, so no rounding step is needed: the enclosure is outward by the algebraic interval rules themselves. Decimal strings such as `"0.1"` and JSON numeric coefficients are rejected; write `"1/10"`. An approximate decimal can suggest a rational center, but the proof is for the rational input actually supplied.

Natural interval evaluation may overestimate a range. For example, evaluating two appearances of the same variable independently loses correlation. This can cause a false negative. It cannot create an accepted false certificate provided the interval arithmetic and validation are implemented correctly. The verifier recomputes the Jacobian enclosure and all bounds; it never trusts a supplied residual, derivative matrix, or claimed answer.

4. Relation to interval Newton, Krawczyk and alpha theory

The mean-value enclosure

$$K = c - YF(c) + (I - YJ)(B - c)$$

contains $T(B)$. This is an interval enclosure, usually larger than the actual image of the map. Our symmetric estimate uses $K \subseteq c - YF(c) + \prod_i [-(Mr)_i, (Mr)_i]$ and checks that latter box lies in B. This is a conservative sufficient contraction test related to Krawczyk methods. It is not a claim to implement every interval-Newton or Krawczyk inclusion theorem. Rump's account gives a mean-value enclosure and separate strict interior inclusion results; stronger conclusions require their stated hypotheses, not merely a reference to the word Krawczyk. Rump (2010), §13, Theorems 13.1–13.3.

The historical methods are classical: see Krawczyk (1969) and Moore (1977). Our contribution here is an educational, inspectable implementation and worked examples, not a new certification theorem.

With $DF(c)$ nonsingular, Smale's alpha theory uses an inverse-Jacobian residual $\beta = \|DF(c)^{-1}F(c)\|$ and higher derivative control γ. Here the cited formulation uses the Euclidean (Hermitian) norm on $\mathbb{C}^n$ and the specified induced higher-derivative operator norms, rather than our weighted maximum norm. For a square polynomial system, the standard bound $\alpha = \beta\gamma < (13 - 3\sqrt{17})/4$ certifies quadratic Newton convergence and distance at most 2β to an associated root. Our bound instead controls one fixed map throughout a specified real box and proves uniqueness throughout that box; its convergence proof gives a geometric rate. Alpha theory's association statement by itself is not a global uniqueness assertion or a uniqueness assertion for an arbitrary input box. The exact rational mode of `alphaCertified` is rigorous; arbitrary precision floating arithmetic without controlled rounding is not automatically so. Hauenstein–Sottile, Theorem 2 and §4.

5. A reusable workflow and its limits

1. Write down the exact rational polynomial system you intend to certify. Check equation order and variable order; a certificate for a mistyped system is still a certificate for the mistyped system.
2. Obtain a candidate center by any method. Numerical root finding is a discovery aid and is outside the verifier's trust boundary.
3. Choose positive rational radii and a rational matrix close to the inverse Jacobian at the center. Unequal radii also choose the weighted norm.
4. Run the verifier, inspect the recomputed Jacobian intervals, residual step, self-map slack and q. Read the error bound only on acceptance.
5. On failure, inspect whether residual scale, derivative variation, interval overestimation or singular behavior is the obstruction. A better center, smaller box, different scaling or different method may help; no remedy is guaranteed. Preserve the rejected output as a diagnostic, not a proof of nonexistence.

EXAMPLES.md gives complete arithmetic for two systems and small-residual failures. SCHEMA.md defines the executable input contract. VERIFICATION.md records tests, review, and trust limits.

The theorem permits general continuously differentiable maps when their derivative enclosures are justified. This executable accepts only rational polynomials, so domain and differentiability are automatic. Transcendental functions, uncertain coefficients, non-square systems, complex roots, automated global root enumeration, singular-root methods, certificate search and continuation are outside this edition. Certification is local to the closed box; other zeros may occur elsewhere. No finite test suite is a formal proof of the Python interpreter or this program. Exact arithmetic, the mathematical proof, source inspection, and adversarial testing play distinct roles.

Hand-checkable examples

All numbers in these certificates are integers or rational numbers. Decimal approximations are unnecessary. The inequalities below use ordinary rational arithmetic and the interval rules in the study.

For a polynomial **map** $F : \mathbb{R}^2 \to \mathbb{R}^2$, the expression $F(c)$ is its vector **value** at the center. The derivative $DF(z)$ is a matrix depending on the point z. The interval matrix $J(X)$ encloses all those derivative values on the closed box $X = \prod_i [c_i - r_i, c_i + r_i]$. It is not a single derivative value.

The verifier forms the fixed-point map $T(z) = z - YF(z)$, encloses its derivative by $A = I - YJ(X)$, and sets

$$M_{ij} = \max(|\inf A_{ij}|, |\sup A_{ij}|), \quad d_i = |(YF(c))_i|, \quad q = \max_i \sum_j M_{ij} \frac{r_j}{r_i}, \quad \eta = \max_i \frac{d_i}{r_i}.$$

The self-map test is the coordinatewise inequality $d + Mr \le r$. The contraction test is the strict inequality $q < 1$. With nonsingular Y, the self-map test supplies existence by the fixed-point argument in the study. The contraction test separately supplies uniqueness. When both hold and Y is nonsingular, the theorem certifies one and only one zero in the **closed box** X, with

$$|z_i^* - c_i| \le \frac{\eta r_i}{1 - q}.$$

Here η is a common maximum: substituting each individual d_i/r_i into that bound is generally unjustified because the coordinates interact. These are local claims. Global root counts stated below follow from separate elementary algebra, not from the certificate.

1. A circle and a line

Certificate: `examples/circle-line.json`.

Use

$$F(x, y) = \begin{pmatrix} x^2 + y^2 - 1 \\ x - y \end{pmatrix}, \quad c = \begin{pmatrix} 7/10 \\ 7/10 \end{pmatrix}, \quad r = \begin{pmatrix} 1/100 \\ 1/100 \end{pmatrix}, \quad X = [69/100, 71/100]^2.$$

Both coordinates are coupled in the first equation. Choose the exact inverse of the center derivative as the preconditioner:

$$DF(c) = \begin{pmatrix} 7/5 & 7/5 \\ 1 & -1 \end{pmatrix}, \quad Y = \begin{pmatrix} 5/14 & 1/2 \\ 5/14 & -1/2 \end{pmatrix}, \quad \det Y = -5/14 \neq 0.$$

The derivative enclosure and center value are

$$J(X) = \begin{pmatrix} [69/50, 71/50] & [69/50, 71/50] \\ 1 & -1 \end{pmatrix}, \qquad F(c) = \begin{pmatrix} -1/50 \\ 0 \end{pmatrix}, \qquad YF(c) = \begin{pmatrix} -1/140 \\ -1/140 \end{pmatrix}.$$

For example, $(YJ)_{11} = (5/14)[69/50, 71/50] + 1/2 = [139/140, 141/140]$. Every entry of $I - YJ(X)$ is consequently $[-1/140, 1/140]$, so

$$M = \begin{pmatrix} 1/140 & 1/140 \\ 1/140 & 1/140 \end{pmatrix}, \qquad q = 1/70, \quad \eta = 5/7.$$

The self-map upper bound in each coordinate is

$$\frac{1}{140} + \frac{1}{140}\frac{1}{100} + \frac{1}{140}\frac{1}{100} = \frac{51}{7000} < \frac{1}{100},$$

with margin $19/7000$ in each coordinate. Thus the box contains a unique zero, and the error bound is

$$|z_i^* - 7/10| \leq \frac{(5/7)(1/100)}{1 - 1/70} = \frac{1}{138} \quad (i = 1, 2).$$

Separately, substitution $x = y$ shows that the two real roots in the whole plane are $(1/\sqrt{2}, 1/\sqrt{2})$ and $(-1/\sqrt{2}, -1/\sqrt{2})$. The negative root lies outside the certified box. The certificate itself did not need these radical expressions.

2. A circle and a parabola

Certificate: `examples/circle-parabola.json`.

Both equations are nonlinear:

$$F(x, y) = \begin{pmatrix} x^2 + y^2 - 1 \\ x^2 - y \end{pmatrix}, \quad c = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}, \quad r = \begin{pmatrix} 1/20 \\ 1/20 \end{pmatrix}, \quad X = [3/4, 17/20] \times [11/20, 13/20].$$

The exact center derivative and its inverse are

$$DF(c) = \begin{pmatrix} 8/5 & 6/5 \\ 8/5 & -1 \end{pmatrix}, \quad Y = \begin{pmatrix} 25/88 & 15/44 \\ 5/11 & -5/11 \end{pmatrix}, \quad \det Y = -25/88 \neq 0.$$

The exact data needed by the checker are

$$J(X) = \begin{pmatrix} [3/2, 17/10] & [11/10, 13/10] \\ [3/2, 17/10] & -1 \end{pmatrix}.$$

$$F(c) = \begin{pmatrix} 0 \\ 1/25 \end{pmatrix}, \quad YF(c) = \begin{pmatrix} 3/220 \\ -1/55 \end{pmatrix}, \quad d = \begin{pmatrix} 3/220 \\ 1/55 \end{pmatrix}.$$

The interval products are worth writing out, since the verifier does not silently cancel dependent occurrences of the same interval:

$$\begin{aligned}
(YJ)_{11} &= \left(\frac{25}{88} + \frac{15}{44}\right)[3/2, 17/10] = [15/16, 17/16], \\
(YJ)_{12} &= \frac{25}{88}[11/10, 13/10] - \frac{15}{44} = [-5/176, 5/176], \\
(YJ)_{21} &= \frac{5}{11}[3/2, 17/10] - \frac{5}{11}[3/2, 17/10] = [-1/11, 1/11], \\
(YJ)_{22} &= \frac{5}{11}[11/10, 13/10] + \frac{5}{11} = [21/22, 23/22].
\end{aligned}$$

In the actual derivative of T, the $(2,1)$ entry is identically zero. The interval matrix computation forgets this dependency and still gives a valid, wider enclosure. Consequently

$$A = \begin{pmatrix} [-1/16, 1/16] & [-5/176, 5/176] \\ [-1/11, 1/11] & [-1/22, 1/22] \end{pmatrix}, \quad M = \begin{pmatrix} 1/16 & 5/176 \\ 1/11 & 1/22 \end{pmatrix}.$$

The row sums are $1/11$ and $3/22$. Equal radii therefore give

$$q = 3/22 < 1, \quad \eta = \max(3/11, 4/11) = 4/11,$$

and the self-map test reads

$$d + Mr = \begin{pmatrix} 3/220 + (1/11)(1/20) \\ 1/55 + (3/22)(1/20) \end{pmatrix} = \begin{pmatrix} 1/55 \\ 1/40 \end{pmatrix} \le \begin{pmatrix} 1/20 \\ 1/20 \end{pmatrix}.$$

The margins are $7/220$ and $1/40$. The error bound is

$$|x^* - 4/5| \le 2/95, \qquad |y^* - 3/5| \le 2/95, \qquad \frac{(4/11)(1/20)}{1 - 3/22} = \frac{2}{95}.$$

Separately, $y = x^2$ gives $t^2 + t - 1 = 0$, where $t = x^2 \ge 0$. Thus the two real roots in the whole plane are

$$\left(\pm\sqrt{\frac{\sqrt{5}-1}{2}}, \frac{\sqrt{5}-1}{2}\right).$$

The certificate locates the positive root and proves uniqueness in its box; it does not assert global uniqueness.

3. Tiny residual, singular center, and two roots

Certificate: examples/tiny-residual-two-roots.json.

Consider

$$F_-(x,y) = \begin{pmatrix} x^2 - 10^{-12} \\ y - x \end{pmatrix}, \quad c = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad r = \begin{pmatrix} 1/1000 \\ 1/1000 \end{pmatrix}, \quad Y = I.$$

Here 10^{-12} means the exact rational $1/1000000000000$; it is not a floating-point datum. The residual is tiny: $F_-(c) = (-10^{-12}, 0)^T$. However,

$$DF_-(c) = \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}$$

is singular. The chosen $Y = I$ is nonsingular, as required by the theorem, but it cannot repair the lack of a contraction on this box. Indeed,

$$J(X) = \begin{pmatrix} [-1/500, 1/500] & 0 \\ -1 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} [499/500, 501/500] & 0 \\ 1 & 0 \end{pmatrix}, \quad M = \begin{pmatrix} 501/500 & 0 \\ 1 & 0 \end{pmatrix}.$$

Thus

$$q = 501/500 > 1, \quad \eta = 1/1000000000, \quad d + Mr = \begin{pmatrix} 1002000001/1000000000000 \\ 1/1000 \end{pmatrix}.$$

The first self-map margin is $-2000001/1000000000000$, so both the self-map and contraction tests fail. The checker reports **not certified**, with no error certificate. There are in fact two roots in the box:

$$(1/1000000, 1/1000000), \qquad (-1/1000000, -1/1000000).$$

The small residual alone gave neither existence nor uniqueness in the specified box.

4. The same tiny residual size, and no real root

Certificate: examples/tiny-residual-no-root.json.

Replace the first equation above by $x^2 + 10^{-12}$, giving

$$F_+(x,y) = \begin{pmatrix} x^2 + 10^{-12} \\ y - x \end{pmatrix}.$$

The same c, r, Y give exactly the same $J(X), A, M, d, q, \eta$ and self-map test as in Example 3: taking the absolute value of the center correction erases the residual's sign. The checker again reports **not certified**. Nevertheless $x^2 + 10^{-12} > 0$ for every real x, so this system has no real root at all. That nonexistence conclusion follows from this separate inequality, not from a failed certificate.

Together Examples 3 and 4 demonstrate the distinction: a rejected candidate certificate is compatible with two roots, no roots, and other possibilities. It is not a decision procedure for root existence.